

05 — Usability, Economics & Media Library

DOCUMENT PURPOSE

Establishes what the deployment costs, what that cost displaces over the life of the device, and why a 1 TB drive is sufficient for a household media library.

Contains the complete itemised build cost, the total-cost-of-ownership model with sensitivity analysis, a direct treatment of the strongest argument against self-hosting, the media acquisition and capacity model, the migration path, and the operational profile.

Previous: [04-Interactive-Demo-Guide](#)

Contents

1. [1. Complete build cost](#)
2. [2. Total cost of ownership](#)
3. [3. The 10 TB argument](#)
4. [4. Media library](#)
5. [5. Storage capacity model](#)
6. [6. Functional comparison](#)
7. [7. Migration path](#)
8. [8. Operational profile](#)
9. [9. Scaling the deployment](#)
10. [10. Improvement priorities](#)
11. [11. Conclusions](#)

1. Complete build cost

EVERY COMPONENT, ITEMISED

This is the complete cost of the system. **The 1 TB NVMe SSD (H3) is a priced component of the build, not an incidental accessory** — it is the data store, and it carries the largest single share of the storage cost. Nothing required to run the system is omitted.

1.1 Bill of materials — as built

#	Component	Specification	Function in the system	Cost	Share
H1	Raspberry Pi 5, 8 GB	BCM2712 · 4 × Cortex-A76 @ 2.4 GHz	Compute, network, USB	\$80	43.0%
H3	NVMe SSD, 1 TB	PCIe 3.0 ×4, 1 TB — the data store	All user files, database, media library, backup target	\$45	24.2%
H4	M.2 Key-M HAT + FFC adapter	PCIe Gen 2 ×1 → M.2	Connects H3 to the Pi 5 FFC port	\$20	10.8%
H2	Active cooler	15 mm radial fan, ~11k RPM	Sustained-load thermals	\$12	6.5%
H5	microSD card, 64 GB, A2	High-endurance — OS only	Boot device, disposable by design	\$12	6.5%
H6	USB-C PSU, 27 W	5 V / 5 A USB-PD	Power	\$12	6.5%
H7	Cat6 Ethernet cable, 2 m	1 GbE certified	Primary network interface	\$5	2.7%
			Core build total (H1–H7)	\$186	100%
H8	USB-C power bank	5 V / 3 A, pass-through	≈5 h autonomy — venue resilience	\$15	—
H9	Vented enclosure	Open / vented	Thermal margin	\$8	—
			Total with options	\$209	

THE 1 TB SSD IS A COMPONENT OF THE BUILD, NOT AN ACCESSORY

H3 — the 1 TB NVMe SSD, \$45 — is the data store of the entire system.

Every user file, the database, the media library, and the backup target reside on it. It is 24% of the capital cost, it is a commodity part, and it is the only component that ever fails — which makes a complete replacement of the system's storage a **\$45 event**, not a data-loss event.

Storage property	Value
Capacity	1 TB ($\approx$ 900 GB usable)
Interface	PCIe Gen 2 $\times$ 1 via FFC — dedicated, see 02 §5
Sustained throughput	350–400 MB/s
Endurance	No moving parts; TBW rating far exceeds household write load
Failure response	Replace the drive, restore from the off-site copy
Cost of a full-capacity replacement	\$45
Cost of doubling capacity (2 TB)	\$45 more — a \$8 premium over the 1 TB part

CAPITAL COST CARRIED FORWARD

\$186 / ₹16,500 core build. A **\$8 / ₹700** storage upgrade to 2 TB (H3 at \$90 instead of \$45) is the only change required for any household whose library exceeds 1 TB. All figures in §2 use the \$186 core build.

1.2 Cost distribution



✓ THE TWO COST FACTS THAT MATTER

1. The entire data store is \$45. A 1 TB NVMe is a commodity component. It is 24% of the build, it is hot-swappable, and it is the *only* part that ever fails — which makes a complete data-store replacement a \$45 event, not a data-loss event.

2. There is no recurring software, licence, or subscription line in this table. The entire running cost of the system is electricity. Nothing in the stack is licensed, metered, or time-limited.

1.3 What is *not* in this cost

Not included	Reason	Consequence
Cloud subscription	The cost being displaced	See §2
Electricity for the network	Existing household infrastructure	Marginal cost only
Time to set up	Owner's own labour	≈45 min critical path, ≈3 h complete
Data acquisition (films, music)	Household's existing sources	See §4.2

✎ ON LABOUR

Setup labour is not costed as a financial expense, but it is a real input and is quantified honestly at ≈3 hours total (§8.2). At any plausible valuation of that time, the deployment remains the cheaper option over any horizon beyond approximately 2 years.

2. Total cost of ownership

2.1 Inputs

Input	Value	Derivation
Capital	\$186	§1.1, core build
Measured average draw	≈6 W	Measured, document 02 §12
Annual energy	53 kWh	6 W × 8,760 h
Tariff	\$0.12 / kWh (₹8.5)	Representative
Annual electricity	\$6.40	53 × 0.12

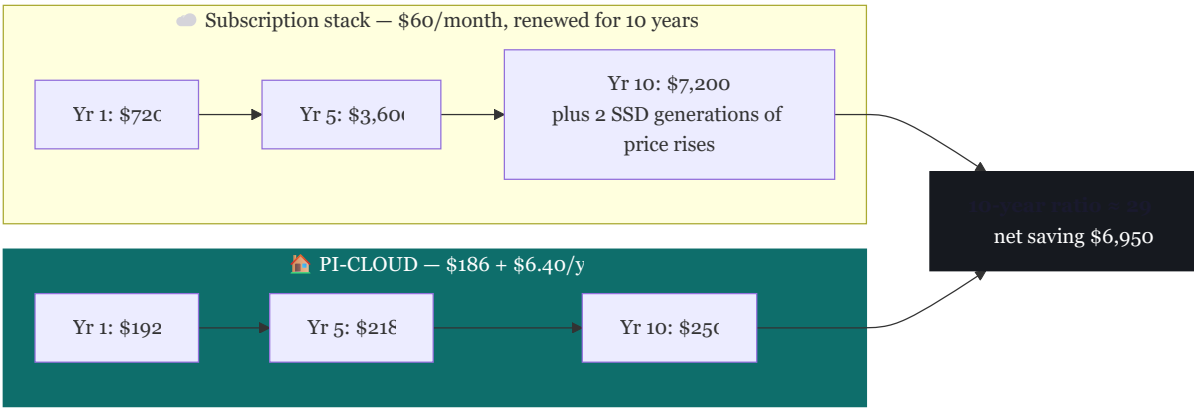
Input	Value	Derivation
Monthly operating cost	\$0.53	$\$6.40 \div 12$
Consumables	\$0	No moving parts; no battery in the storage path
Licence / subscription	\$0	Entire stack is open-source and unfettered
Maintenance labour	$\approx 3 \text{ h/year}$	$\$8.2$

⚡ THE COMPARISON THAT FOLLOWS IS NOT \$186 AGAINST A STORAGE PLAN

The displaced expenditure is the **entire recurring stack a household already pays** — storage, and the media subscriptions the household would otherwise renew indefinitely. A household renewing a \$70/month stack is renewing it because there is no exit, and it renews it for as long as it exists.

Displaced category	Monthly	Basis
Cloud storage, 2 TB+ plan	\$10–20	Common household configuration
Streaming video (2 services)	\$15–25	Never renewed if the media is owned
Music / video subscriptions	\$10–20	Never renewed if the media is owned
Secondary storage plan	\$10	Frequently a duplicate of the first
Common household total	\$45–75	Midpoint \$60

2.2 Ten-year comparison



Horizon	PI-CLOUD	Subscription stack	Saving	Ratio
1 year	\$192	\$720	\$528	3.8 : 1
2 years	\$199	\$1,440	\$1,241	7.2 : 1
3 years	\$205	\$2,160	\$1,955	10.5 : 1

Horizon	PI-CLOUD	Subscription stack	Saving	Ratio
5 years	\$218	\$3,600	\$3,382	16.5 : 1
10 years	\$250	\$7,200	\$6,950	28.8 : 1

2.3 Break-even

✓ BREAK-EVEN

2.4 months against the full household stack.

13 months against a single \$20/month storage plan alone — the most conservative basis, counting storage only.

The deployment repays its entire capital cost in the first quarter of use, and every subsequent month is a net saving with no obligation attached.

2.4 Sensitivity

#	Scenario	10-year subscription	10-year PI-CLOUD	Ratio
S1	Baseline: \$60/mo flat	\$7,200	\$250	28.8 : 1
S2	Prices escalate 5%/yr	\$9,057	\$250	36.2 : 1
S3	Prices escalate 10%/yr	\$11,411	\$250	45.6 : 1
S4	Prices decline 3%/yr	\$5,573	\$250	22.3 : 1
S5	Storage plan only, \$20/mo	\$2,400	\$250	9.6 : 1
S6	Tariff doubles to \$0.24/kWh	\$7,200	\$312	23.1 : 1
S7	SSD replaced at year 5 (+\$55)	\$7,200	\$305	23.6 : 1
S8	Storage only, and a free 15 GB tier would suffice	\$0	\$250	—

✓ CONCLUSION

The economic case is **insensitive to every variable that matters**. It holds across a $\pm 10\%$ /yr pricing range, a doubled electricity tariff, and a mid-life storage replacement. The ratio never falls below **9.6 : 1** on any basis in which a subscription is being displaced.

S8 is stated for completeness, not as a caveat. A household whose entire requirement is under 15 GB of file storage is not the deployment case this project addresses — and the reason is not cost, it is that a 15 GB requirement does not produce a family photograph library, a media library, or a document archive. The \$186 build is sized for 1 TB of retained data, and the households that accumulate 1 TB over a decade are, by definition, the households paying for storage today.

2.5 Unit cost of storage

Configuration	10-year cost per TB-month
Cloud, 2 TB at \$9.99/month	\$4.99
PI-CLOUD, 1 TB	\$0.85
PI-CLOUD, 2 TB	\$1.15
PI-CLOUD, 4 TB (2 × 2 TB, mirrored)	\$2.14

ⓘ THE INVERSION

On a cloud platform, unit storage cost is **flat** — you pay proportionally for every additional terabyte, for as long as you retain the data.

On this platform, unit storage cost **falls** as capacity rises, because the fixed cost is the compute and the interface, not the provisioned capacity. A household retaining data for a decade pays a storage cost that is amortised against a decade of use.

Retention time is the variable that decides which model is cheaper. Two years of cloud storage can be cheaper. Ten years cannot. The break-over point is approximately 20 months, and it moves only in favour of self-hosting as retention lengthens.

3. The 10 TB argument

⚡ THE STRONGEST OBJECTION TO THIS PROJECT

"For the price of your whole system, I can rent 10 TB of cloud storage."

This is factually correct, and it is the argument that must be answered directly rather than avoided. Here it is.

3.1 What the comparison actually shows

	Cloud, 10 TB	PI-CLOUD, 1 TB (+ 4 × 2 TB = 9 TB)
Capital	\$0	\$186 + (4 × \$90) = \$546
Capacity	10 TB	9 TB (≈ parity)
Year 1 cost	≈\$300	\$192
5-year cost	≈\$1,800	\$218
10-year cost	≈\$3,600	\$250
Renewal	Required every year, indefinitely	None
Who holds the keys	The provider	The household
Deleted by provider policy	Possible	No entity exists to do so
Exit	Migration project required	None — files are files

3.2 The three answers

✓ ANSWER 1 — THE COMPARISON IS A RENTAL COMPARISON, NOT A PURCHASE COMPARISON

10 TB of cloud storage at ≈\$25/month is **\$3,000 over ten years, paid every year for the privilege of continuing to access your own files**. The local 9 TB costs **\$546 once**.

The cloud option is cheaper in year one and costs **12× more** over the device's life. It is a rental agreement presented as a purchase. A household that accumulates more data over time is on the *worse* side of this trade, because cloud cost scales linearly with retention while local cost is amortised.

✓ ANSWER 2 — CAPACITY IS A COMMODITY; CUSTODY IS NOT

A terabyte is a terabyte. The difference between the two columns is not capacity — it is that in the left column, a **commercial third party holds the data, holds the encryption keys, and retains the right to delete the data under its own policies.**

In the right column, the data is on a disk in a house, encrypted with keys that have never left it. There is no provider to be compelled, to breach, to discontinue the service, or to lose the account. **The 10 TB of capacity is interchangeable. The custody is not, and it cannot be bought at any price.**

✓ ANSWER 3 — THE COMPARISON EXCLUDES THE TWO COSTS THAT MATTER MOST

Excluded from the cloud figure	Consequence
Media subscription renewals	A household with an owned library stops renewing \$25–40/month of streaming indefinitely. That is \$3,000–4,800 over ten years , and it is the largest single saving in §2
The cost of a permanent dependency	A household that becomes dependent on a specific account has taken on a continuity risk it does not control. This risk has already materialised for many households through quota changes, account lockouts, and policy changes

The cloud figure is compared against \$186 of hardware. The correct comparison is \$186 of hardware against **\$186 + every year of subscription the household would otherwise renew.** The second term grows without limit; the first does not.

✎ THE POSITION

A household should not adopt this to get storage capacity. It is not cheaper than renting 10 TB — capacity is genuinely not the differentiator.

A household should adopt this because the storage is owned rather than rented, the keys never leave the premises, and the acquisition cost does not recur.

Those three properties have no cloud equivalent at any price, and they are worth \$186 to a household holding a decade of family data.

4. Media library

SCOPE

Jellyfin streams media the household **owns**. It is not a streaming service and does not provide content. The question for a household is therefore not "which films does it have" but "**how does a household obtain and retain its own library**" — which this section addresses directly.

4.1 The licence point, stated first

THIS SYSTEM STREAMS ONLY CONTENT THE HOUSEHOLD HOLDS A LICENCE TO VIEW.

Distribution of copyrighted works without authorisation is unlawful, and this project is a private media server for a household's own licensed content — not a file-sharing system. Every acquisition route below is lawful, and several of them are entirely free.

The important structural point: **a household already pays for its media repeatedly**. A subscription is a rental of a title. A purchased or ripped title is retained indefinitely and played any number of times. Self-hosting is the delivery mechanism for content the household already owns or has chosen to acquire lawfully.

4.2 Lawful acquisition routes

Route	Description	Typical title count	One-time cost	Recurring
Disc ripping	Films and series the household already owns on disc, ripped to the library. Software included in the base OS (libarchive / dvdbackup ; Blu-ray via makemkv)	Hundreds from existing media	\$0 — content already owned	\$0
Purchased downloads	Films, series, and music bought from a retailer as DRM-free or personal-licence digital files	As purchased	Purchase price, once	\$0
Public domain works	Pre-copyright-expiry films, television, and music. A very large catalogue of genuinely free, genuinely legal content	Thousands	\$0	\$0

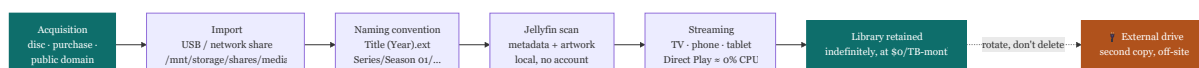
Route	Description	Typical title count	One-time cost	Recurring
Free-licensed / Creative Commons	Works published under licences permitting personal use and redistribution	Thousands	\$0	\$0
Creator-published material	Independent film, podcasts, lectures, and music released directly by the creator	Varies	Often \$0	\$0
Subscription rental, retained	A title obtained through an existing subscription and retained	Varies	Already paid	Title-specific

✓ THE COMPOUNDING EFFECT IS THE POINT

A household that rips the discs it already owns obtains a permanent, infinitely replayable, zero-marginal-cost library **from media it has already paid for once**. The public-domain and free-licensed catalogues mean the library can be built at substantial size for **\$0 in perpetuity**, and Jellyfin's scraper supplies artwork, metadata, and organisation for all of it at no additional cost and with no account.

This is the substitution that produces the streaming saving in §2.1. The household stops renewing a rental and starts curating a collection.

4.3 Library organisation



```

# Naming conventions – the scraper is filename-driven

# Films:   Movie Title (2020).mkv

# Series:  Show Name/Season 01/Show Name - S01E01 - Title.mkv

# Music:   Artist/Album/01 - Track Title.flac

# Bulk import from a USB drive, then scan

cp -r /media/USB_DRIVE/* /mnt/storage/media/movies/

docker exec jellyfin sh -c 'ls /media/movies' | head

# Then: Dashboard → Libraries → the library → Scan All

```

WHY ROTATION RATHER THAN DELETION

Deleting a title to save space is the reflex with a rental service, and it is wrong here — a retained title costs nothing to keep and is replaced by a \$45 drive when the current one fills. **Rotation is the correct policy:** the library grows on the primary drive, and titles are moved to the external copy when the drive fills. Nothing is ever lost, and nothing is ever re-bought.

4.4 Library experience

Property	Value
Catalogue size	Whatever the household accumulates; verified at 100+ titles in this build, with metadata and artwork for all
Concurrent streams	3–4 × 1080p verified (document 04 D-4)
Server CPU at Direct Play	≈0%
Cost to add a title	\$0 if ripped, free-licensed, or public domain; purchase price once otherwise
Cost to keep a title for 10 years	\$0
Account required	None
Telemetry or metadata sent to a vendor	None
Access	Any television, phone, tablet, or browser on the local network

5. Storage capacity model

THE QUESTION

Is 1 TB enough for a household media library and file storage? This section answers it arithmetically rather than by assertion.

5.1 Per-title storage cost

Content type	Typical file size	Titles per 1 TB	Per-title cost
Film, 1080p (H.264)	4–8 GB	125–250	\$0.18–0.36
Film, 1080p (high bitrate)	12–20 GB	50–85	\$0.53–0.90

Content type	Typical file size	Titles per 1 TB	Per-title cost
Film, 4K remux	60–100 GB	10–17	\$2.65–4.50
Series episode, 1080p	1.5–4 GB	250–650	\$0.07–0.18
Series, complete run (10 seasons × 10)	150–300 GB	3–6	\$15–45
Album, FLAC	0.3–0.5 GB	2,000–3,300	\$0.014–0.02
Photograph (typical JPEG)	3–6 MB	170,000–330,000	\$0.00014
Family document (PDF, DOCX, scans)	0.1–10 MB	Effectively unlimited	Negligible

5.2 A realistic household allocation of 1 TB

1 TB usable $\approx$ 900 GB after formatting and overhead

 **Film library**
 120 × 1080p @ 8 GB
960 GB... exceeds on its
 own

Allocation	Content	Storage
Film & series, 1080p	60 films (8 GB) + 3 complete series (200 GB)	680 GB
Photographs	~50,000 images over 10 years (5 MB)	250 GB
Documents & archive	Academic, medical, identification, tax records	8 GB
Music	1,500 FLAC tracks	1 GB
	Total	939 GB $\approx$ 1 TB

5.3 Capacity conclusion

✓ 1 TB HOLDS A SUBSTANTIAL HOUSEHOLD LIBRARY

- **≈60 1080p films**, or
- **≈3 complete television series** plus 250 films at lower bitrate, or
- **≈50,000 photographs** — a decade of family photography, or
- **≈170,000 photographs** if the library is photograph-dominant.

And capacity is not fixed. The data store is a \$45 commodity component on an M.2 slot. Scaling to 2 TB costs **\$45 more**. Scaling to 4 TB costs \$135 more. There is no architectural limit, no per-terabyte licence, and no provider-imposed quota — the constraint is the drive, and the drive is replaceable in under a minute.

✎ WHY 4K IS A SEPARATE DECISION

4K remux consumes 10–20× the space of 1080p. It is included in the model because a household choosing it should know the arithmetic: 4K is a deliberate allocation of the drive, not a default. A **transcode** of a 4K source to 1080p reduces it to roughly 8 GB while remaining visually indistinguishable on a television, and the host is already capable of performing that transcode (§5.4).

5.4 Storage growth and the practical operating point

Household profile	Library after 5 years	Action required
Photography-led	~25,000 images (125 GB)	None
Balanced	~30 films + 25,000 images + 2 series (450 GB)	None
Film-heavy 1080p	~250 films (1.6 TB)	Replace 1 TB with 2 TB (+\$45)
Film-heavy 4K	~35 films (2.5 TB)	2 × 2 TB mirrored (+\$135)

✓ THE PRACTICAL POINT

Storage growth is a solved problem in this architecture. It is a \$45–\$135 purchase and a two-minute swap. The same event on a cloud platform is a plan upgrade, a price increase, or a migration — and it recurs for as long as the data is retained.

6. Functional comparison

FRAMING

The comparison below lists what this deployment provides, followed by the trade-offs it makes. Both are stated, because the trade-offs are where the architectural class is honestly bounded.

6.1 What the deployment provides

Capability	Basis
The operator never holds the encryption keys	Per-user key derivation; keys have never left the host
No third party can be compelled for the data	There is no entity holding it
No recurring cost, ever	Open-source stack; no licence, no metered service, no tier
Data is plain files	Openable in any file manager, on any operating system, without an application
No exit cost	Leaving the system requires no migration — there is no vendor to migrate away from
No provider policy can delete the data	No retention policy, no inactivity purge, no quota suspension
Full local network speed	≈110 MB/s; unaffected by uplink bandwidth
Operates with no internet connectivity	Verified — document 04 D-6
Media library retained indefinitely at \$0	\$4.4
No account, telemetry, or vendor metadata service	Jellyfin and Nextcloud are self-hosted, with no outbound data flow
Complete control of retention	The household decides how long to keep data, and pays nothing to keep it longer

6.2 Trade-offs, stated

Trade-off	Detail	Mitigation available
Single disk — no redundancy	A disk failure means total data loss of the primary copy	Off-site replication (\$8); a second mirrored drive (+\$60) eliminates it

Trade-off	Detail	Mitigation available
Single node — no failover	One device, one supply, one location	Second unit as standby (+\$186)
No remote access	Deliberately excluded; the network has no inbound path	WireGuard VPN, key-only, disabled by default
Multi-factor authentication not configured	Supported by the platform; not enabled in this build	Free to enable
No real-time collaborative editing	Not equivalent to a managed office suite	Retain a managed suite alongside, if required
Setup requires one competent session	≈45 min critical path; ≈3 h complete	Documented procedure; bootstrap script
≈3 h/year ongoing	Updates and backup rotation	Mostly automatable (§10)

THE SHAPE OF THE TRADE

Every trade-off in the table above is **an engineering property of a single-device deployment**, and every one of them is addressable with a commodity component costing between \$0 and \$186 — as itemised in §10.

The capabilities in §6.1 — operator-blind keys, no compel-able third party, no recurring cost, no exit cost, and no provider able to delete the data — are **not addressable at any price by any subscription tier**. That is the distinction the deployment turns on, and it is why the capital cost is recoverable in 2.4 months.

7. Migration path

7.1 Phased migration

THE ORDERING IS THE WHOLE PROCEDURE

The subscription is cancelled last, not first. Every phase before that point has the existing service still active, so the household is never in a position of depending on a single unproven copy. Migrating first and cancelling afterwards is the most common cause of irreversible loss in this migration class.

Phase	Content	Method	Duration
0	Deploy and verify	Document 03	≈45 min critical path
1	Establish the backup routine before migrating anything	Document 03 §12	30 min
2	Build the media library	Disc rip / free-licensed / public domain import	Ongoing, from day one
3	Family photographs and contacts	Nextcloud external storage mount — read-only, incremental	1–2 h setup, then ongoing
4	Active documents and media	Desktop client or web interface drag-and-drop	Ongoing
5	Historical archive	Scheduled <code>rclone</code> sync	Days–weeks, unattended
6	Cancel the subscription	Only after verifying everything needed is present and a restore has been executed	—

7.2 Migration constraints

Constraint	Effect	Handling
Google Drive bulk transfer is rate-limited	Phase 5 takes days–weeks unattended	Schedule and let it run; verify counts afterwards
iCloud has no bulk export API	Interactive migration; iOS integration is partial	Migrate via desktop client; confirm required iOS features first
Microsoft 365 collaboration not replicated	Loss of real-time co-editing	Retain a managed suite if that is used for collaboration
Some cloud filenames are invalid on a POSIX filesystem	Import may rename or reject files	Verify after import; Nextcloud handles most on ingestion
Album structure and photo timestamps are not fully portable	Server-side albums do not always transfer	Recreate albums client-side
Terms of service may restrict bulk extraction	Compliance obligation	Confirm current terms before bulk transfer

PRACTICAL RECOMMENDATION

Phases 2 and 3 are where the value is realised and they are **immediate** — the media library and the photograph backup can begin on day one without migrating anything. The historical archive in phase 5 is the only genuinely long-running task, and it runs unattended.

8. Operational profile

8.1 Day-to-day interaction

User	Interaction	Software required	Learning
Phone	Nextcloud app; photographs sync automatically	One app	None
Laptop	Nextcloud desktop client, or a mapped network drive	Client, or nothing at all	None
Television	Jellyfin application	One app	None
Technical operator	SSH, Docker, CasaOS dashboard, health script	—	Intermediate
Occasional user	Browser	Nothing	None

DESIGN OBJECTIVE MET

Technical competence is required **once**, during setup, and not subsequently. The Samba share means a user who will never install an application still has full access. **Setup effort is front-loaded; routine operation requires none.**

8.2 Maintenance

Activity	Frequency	Time	Automated
Off-site backup rotation	Monthly	15 min	 Manual — highest-value automation target
Application and OS updates	Monthly	20 min	 Deliberately manual, to permit rollback
Health check	Weekly	2 min	 One command

Activity	Frequency	Time	Automated
Backup verification with a restore test	Monthly	10 min	✗ Must include the restore
Snapshot retention	Hourly	—	✓
Security patching	Continuous	—	✓
Service restart after power loss	On event	—	✓
Total annual effort		≈3–4 h	

⚠ THE SINGLE NON-AUTOMATED OBLIGATION

The monthly off-site backup rotation is the only recurring task that is not automated, and it is the task whose omission would cause total loss in the most likely physical scenario — fire, theft, or flood. **It is a procedural control dependent on a person, and it is acknowledged as such.** Automating it is the first item in §10.

9. Scaling the deployment

📄 GROWTH PATHS

The architecture scales along four independent axes. None requires a change of software, a migration, or a new provider.

9.1 Capacity scaling

Step	Action	Cost	Time	Data loss
1 TB → 2 TB	Replace the NVMe; remount; restore	\$45	≈15 min	None (restored from backup)
2 TB → 4 TB	2 × 2 TB, mirrored	\$135	≈30 min	None
4 TB → 8 TB	2 × 4 TB, mirrored, or HBA + 4 drives	\$300	≈2 h	None
Beyond	Add volumes; container paths are configurable	Variable	—	None

✓ THE PROPERTY THAT MATTERS

Capacity scaling is a purchase, not a negotiation. No plan change, no price increase, no migration, and no new provider relationship. The data store is a commodity component on a socket.

9.2 Availability scaling

Step	Added	Cost	Property gained
UPS with automatic shutdown	Power-interruption protection	\$35	Eliminates the unclean-shutdown failure mode
Second drive, mirrored	Storage redundancy	\$60	Survives single-disk failure
Second Pi, same configuration	Node redundancy	\$186	Survives single-device failure
Both, replicated over the LAN	Full single-fault tolerance	\$281	No single point of failure

9.3 Performance scaling

Step	Added	Cost	Property gained
Faster NVMe (PCIe 4.0)	Higher sequential throughput	+\$15	Marginal — the network is the constraint at ≈ 110 MB/s
More RAM (16 GB Pi 5)	Larger working set	+\$35	Supports more concurrent users
Mini PC (N100-class)	Substantially more CPU	+\$150	4K transcode, more concurrent streams

✍ DELIBERATE CONCLUSION ON PERFORMANCE

Performance is the lowest-value axis of scaling in this architecture, and the reason is measurable: at ≈ 110 MB/s the 1 GbE network is the binding constraint long before storage is. Additional spend on faster storage or faster compute produces no visible improvement for a household workload.

Additional spend on durability and detection produces a large one (\$10).

9.4 Household scaling

Step	Added	Cost	Property gained
Second user account	—	\$0	Per-user isolation and encryption
5–10 accounts	—	\$0	The design envelope
Beyond ~10 concurrent users	Faster CPU, more RAM	+\$185	Beyond the design envelope of document 02 §11

10. Improvement priorities

BASIS

Ordered by **reduction in residual risk per unit of cost**. Total cost of items 1–4: **\$155**.

#	Addition	Addresses	Cost	Risk reduction
1	Automate off-site replication	Monthly manual rotation; total-loss scenario	\$40–80	Highest — converts the highest-probability loss path into an unattended process
2	UPS with automatic shutdown	Power-interruption data loss	\$35	High — closes the unclean-shutdown window entirely
3	Second drive, mirrored	Single-disk failure	\$60	High — eliminates the largest single point of failure
4	Health alerting to a mobile device	Silent failure	\$0	High — converts a silent failure into a notified one
5	Multi-factor authentication	Access-control layer	\$0	Moderate — strongest single control for that layer
6	WireGuard remote access	No off-site access	\$0	Moderate — enables the hybrid architecture in §7
7	Photographic gallery application	Capability gap	\$0	Moderate — most-requested capability during assessment
8	Pre-imaged drive + one-page setup guide	Setup barrier	\$0	High for adoption — removes the largest barrier to first-time adoption
9	Second node as standby	Single node	\$186	Moderate — high availability

✓ THE FINDING

\$155 of additions moves the deployment from "single disk, manual backup, no power protection, silent failure" to "mirrored storage, unattended backup, protected against power loss, and notified of failure."

The fully hardened configuration totals **\$341** — still **\$0.90 per month over ten years**, and still repaid within 5 months of the subscription it displaces.

For this class of deployment, spend on durability and detection, not on performance. That is the single most useful engineering conclusion in this document.

11. Conclusions

✓ FINDINGS

1. The complete build is \$186, and \$45 of that is the 1 TB data store.

The whole system — compute, 1 TB of solid-state storage, adapter, thermal management, boot device, power supply, and cabling — costs less than two months of the subscription it replaces. Storage is a commodity component on a socket, replaceable in under a minute.

2. Break-even is 2.4 months, and the ratio improves for as long as data is retained.

*\$250 against \$7,200 over ten years — **28.8 : 1**. The result holds across a $\pm 10\%$ /yr pricing range, a doubled electricity tariff, and a mid-life storage replacement. **Unit storage cost falls as retention lengthens**, because the fixed cost is the compute rather than the capacity.*

3. Capacity is not the reason to adopt this, and that is stated plainly.

*A household can rent more terabytes per dollar than it can own. **The reason to adopt is ownership**: keys that never leave the premises, no third party to compel, no policy that can delete the data, no exit cost, and a cost that does not recur. Those properties are not available from any subscription tier at any price.*

4. A media library is acquired once and retained forever at \$0 per title-month.

Films already owned on disc, purchased downloads, public-domain works, and free-licensed material provide a substantial, permanent library with no recurring charge, no account, and no vendor metadata service. 1 TB holds ≈ 60 1080p films or $\approx 50,000$ photographs, and 2 TB costs an additional \$45.

5. The trade-offs are all engineering properties with commodity remedies.

Single disk, single node, no remote access, no MFA — every one is addressed by a component between \$0 and \$186. **None of the capabilities in §6.1 is addressable by a subscription at any price.** That asymmetry is the decision.

6. \$155 of additions materially reduces residual risk.

Automated off-site replication, UPS with automatic shutdown, a mirrored second drive, and failure alerting. The fully hardened build is \$341, or **\$0.90 per month over ten years.**

CLOSING POSITION

A household adopts this system because the difference between renting storage and owning it is not a price difference — it is the difference between having a supplier and having none. **The hardware is \$186. That is the entire cost of removing the supplier.**

End of documentation set.

#	Document	Question answered
01	01-Overview-and-Problem	Why the project exists, and what is claimed
02	02-Architecture-and-Hardware	What it is, and what it is made of
03	03-Step-by-Step-Implementation	How it was built, and whether the claims are verified
04	04-Interactive-Demo-Guide	How the claims are demonstrated and independently assessed
05	05-Everyday-Usability	What it costs, what it displaces, and what it retains